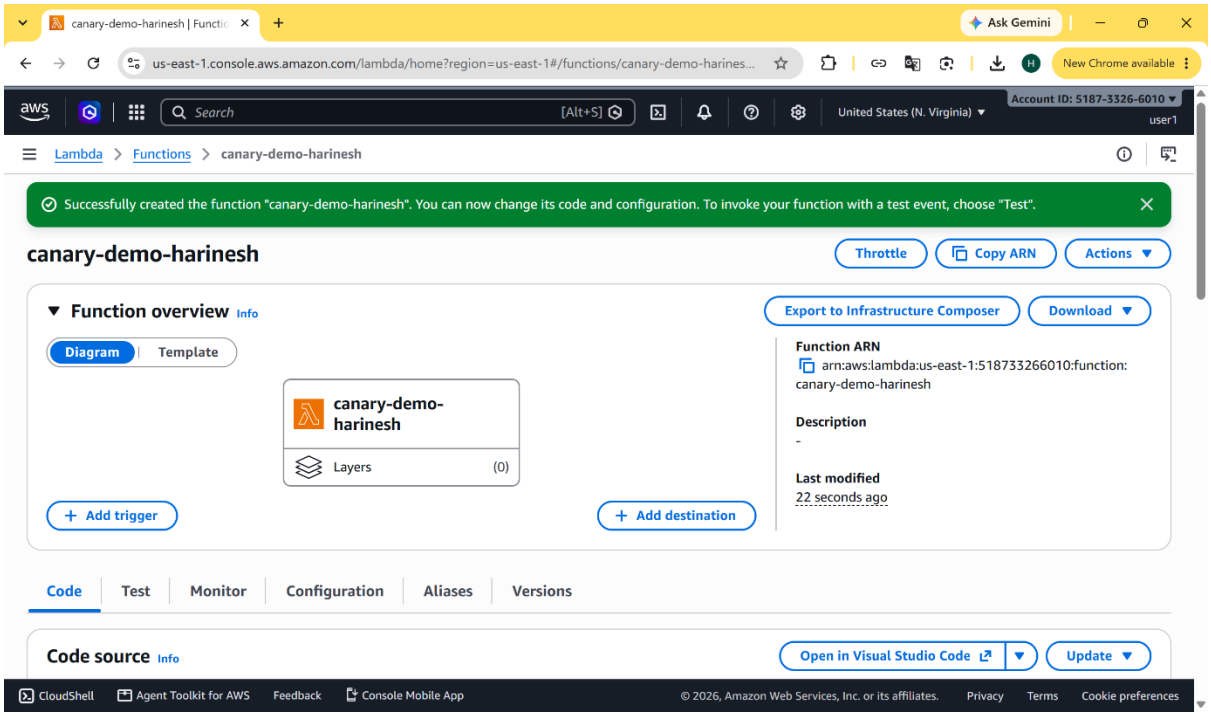
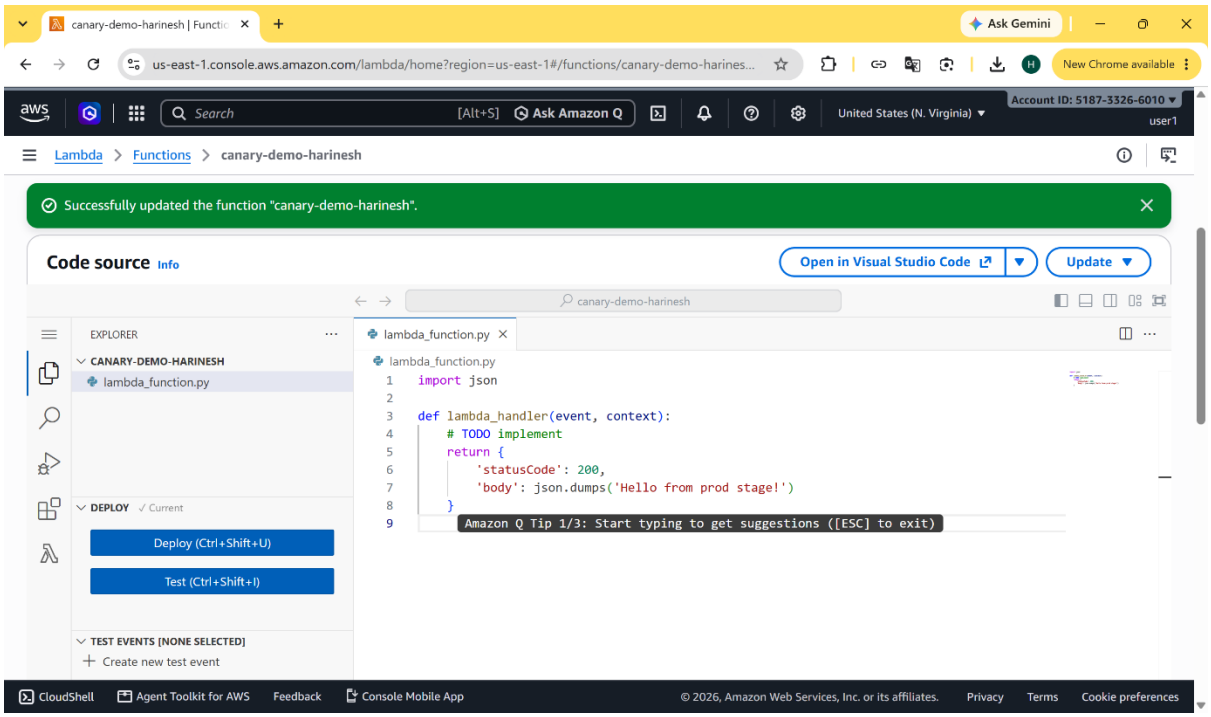
AWS Capstone Project 4

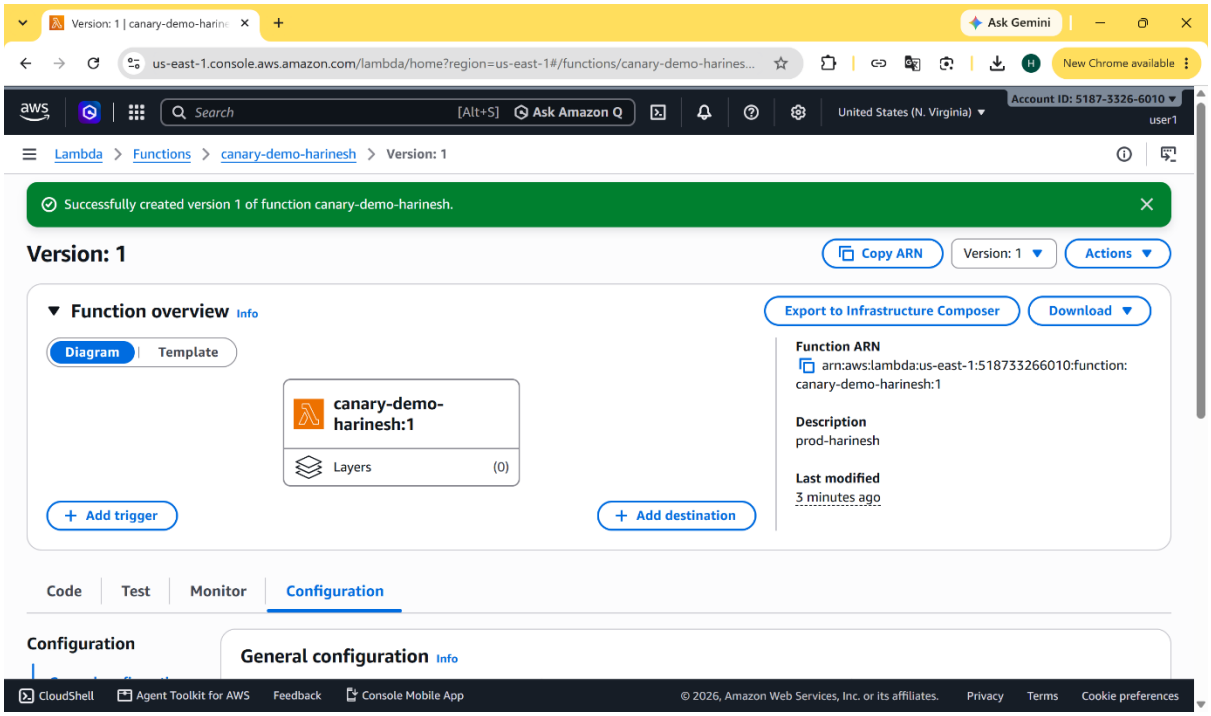
Canary Deployment Using Lambda and API Gateway

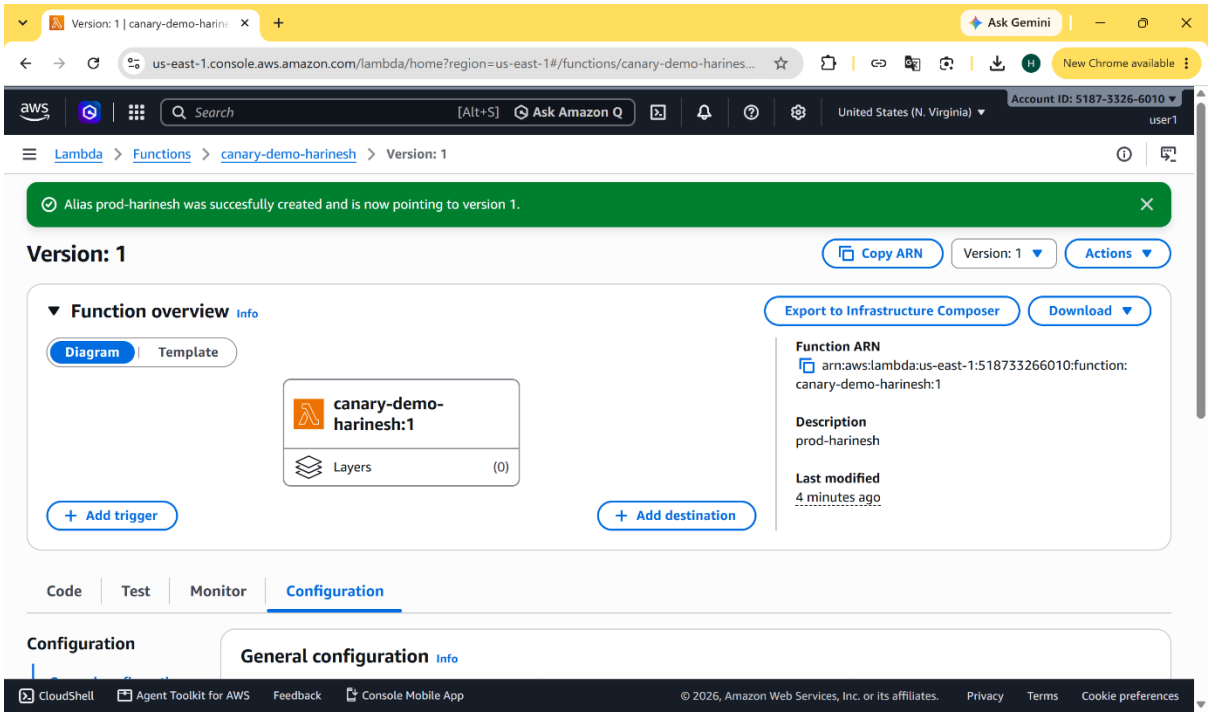
Solution / Screenshots

Task 1: Create a Lambda function:

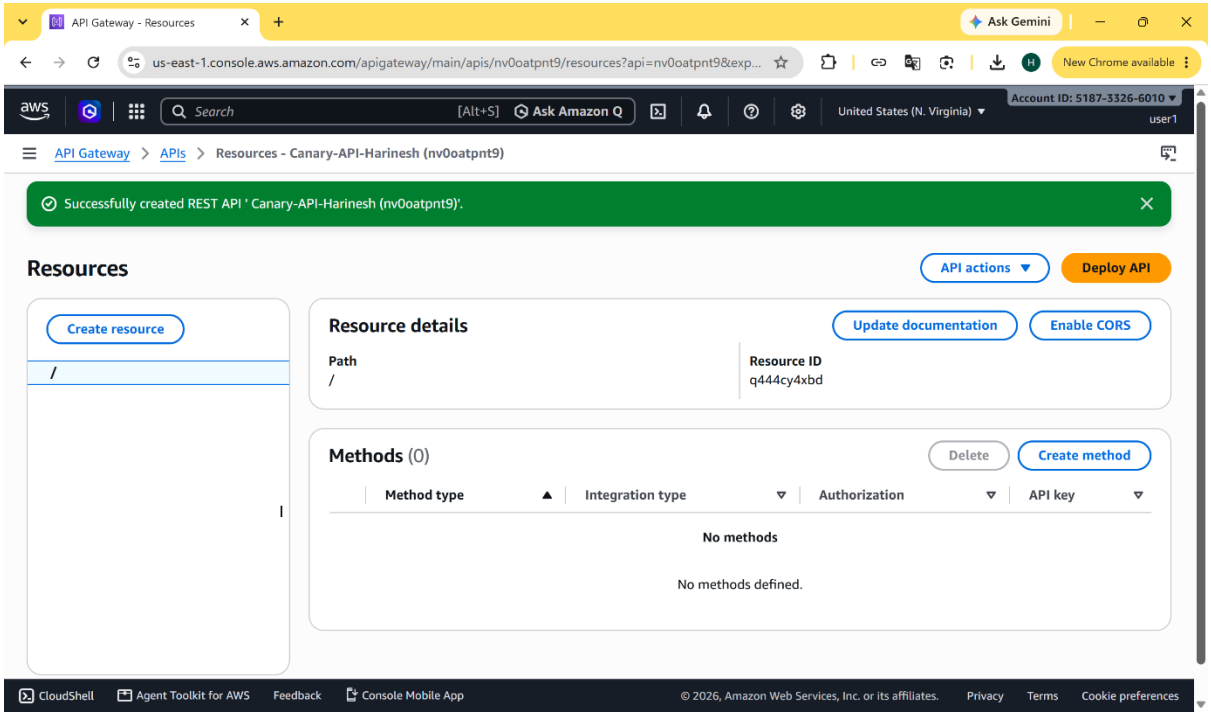


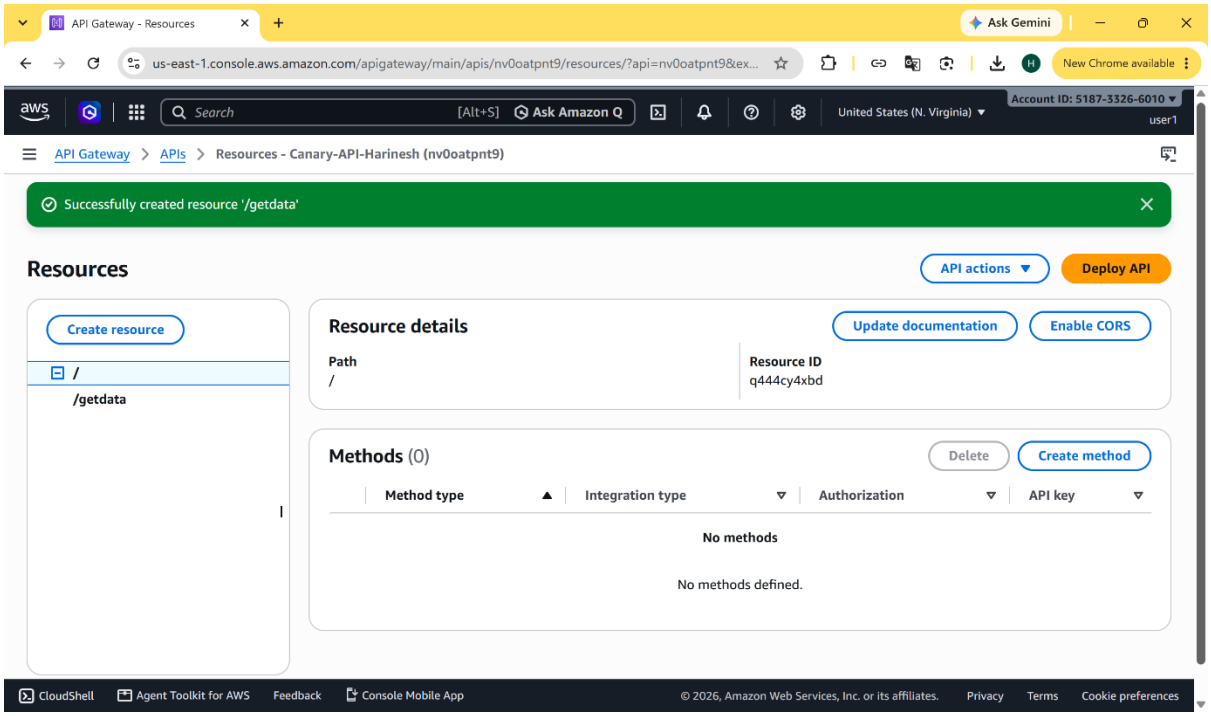


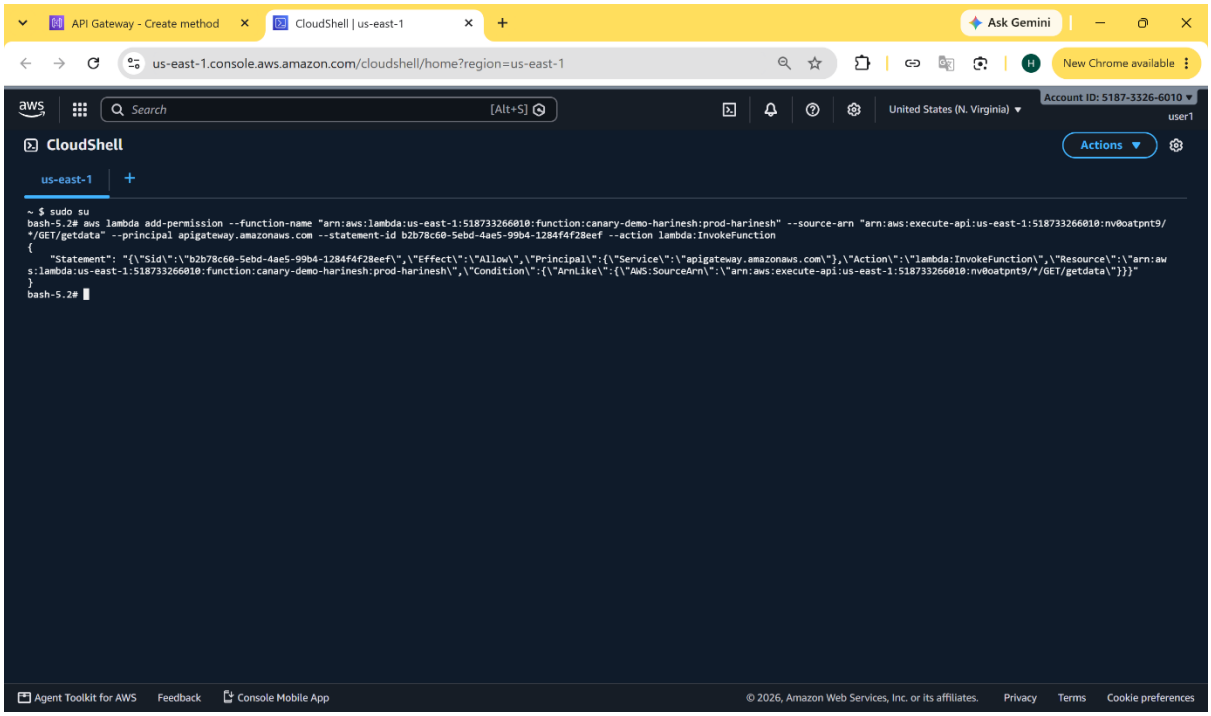

Version: 1 | canary-demo-harine
Ask Gemini
us-east-1.console.aws.amazon.com/lambda/home?region=us-east-1#/functions/canary-demo-harines...
New Chrome available
Search
[Alt+S]
Ask Amazon Q
United States (N. Virginia)
Account ID: 5187-3326-6010
user1
Lambda > Functions > canary-demo-harinesh > Version: 1
Successfully created version 1 of function canary-demo-harinesh.
Version: 1
Copy ARN
Version: 1
Actions
Function overview Info
Export to Infrastructure Composer
Download
Diagram
Template
canary-demo-harinesh:1
Layers (0)
Add trigger
Add destination
Function ARN
arn:aws:lambda:us-east-1:518733266010:function:canary-demo-harinesh:1
Description
prod-harinesh
Last modified
3 minutes ago
Code
Test
Monitor
Configuration
Configuration
General configuration Info
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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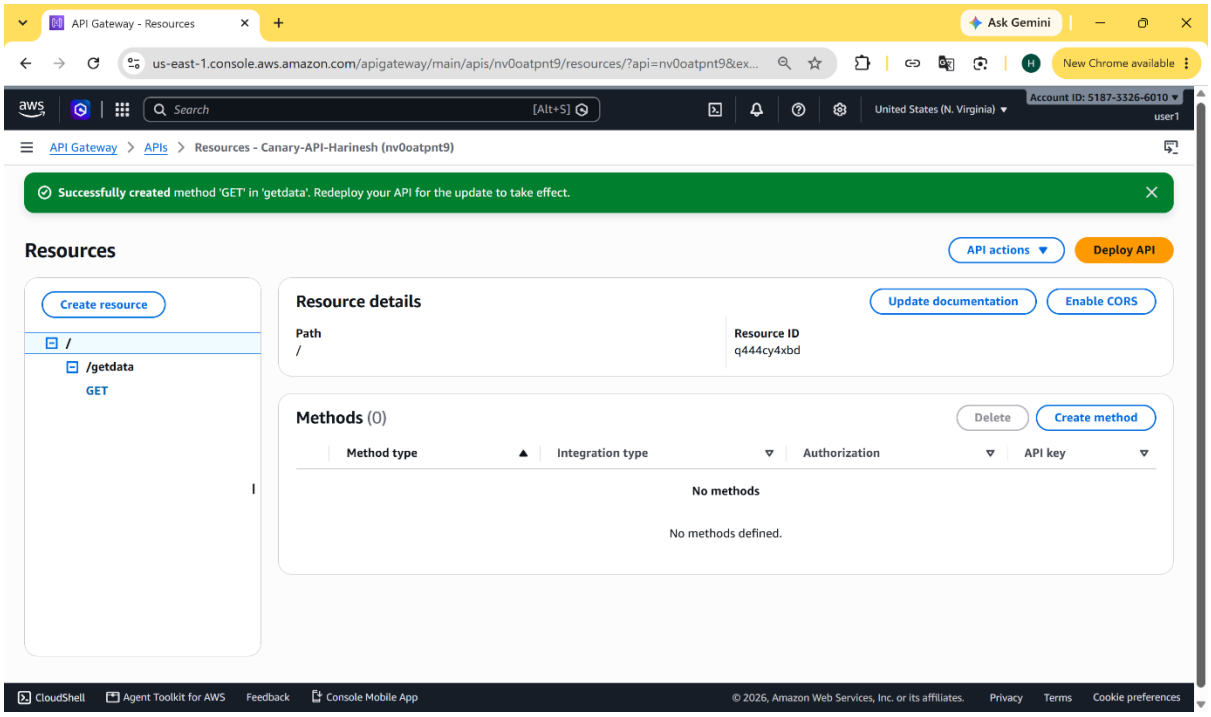

Version: 1 | canary-demo-harine
Ask Gemini
us-east-1.console.aws.amazon.com/lambda/home?region=us-east-1#/functions/canary-demo-harines...
New Chrome available
Search
[Alt+S]
Ask Amazon Q
United States (N. Virginia)
Account ID: 5187-3326-6010
user1
Lambda > Functions > canary-demo-harinesh > Version: 1
Alias prod-harinesh was succesfully created and is now pointing to version 1.
Version: 1
Copy ARN
Version: 1
Actions
Function overview Info
Export to Infrastructure Composer
Download
Diagram
Template
canary-demo-harinesh:1
Layers (0)
Add trigger
Add destination
Function ARN
arn:aws:lambda:us-east-1:518733266010:function:canary-demo-harinesh:1
Description
prod-harinesh
Last modified
4 minutes ago
Code
Test
Monitor
Configuration
Configuration
General configuration Info
CloudShell
Agent Toolkit for AWS
Feedback
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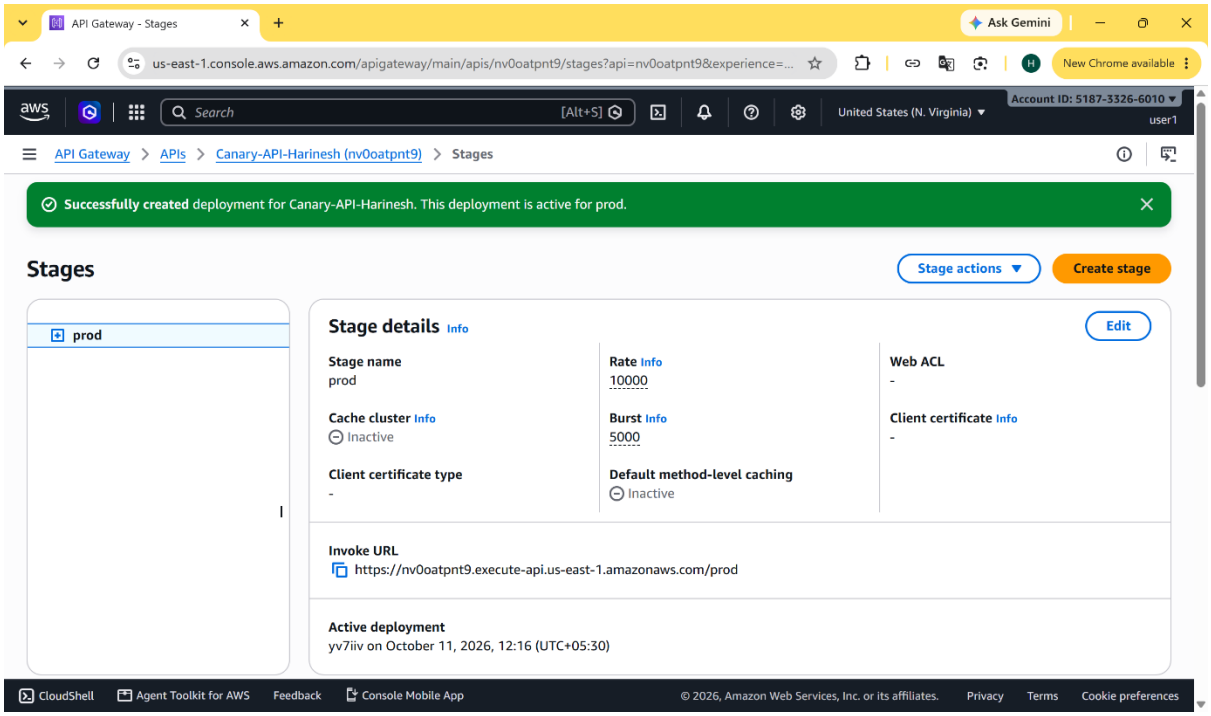
Task 2: Create an API Gateway:

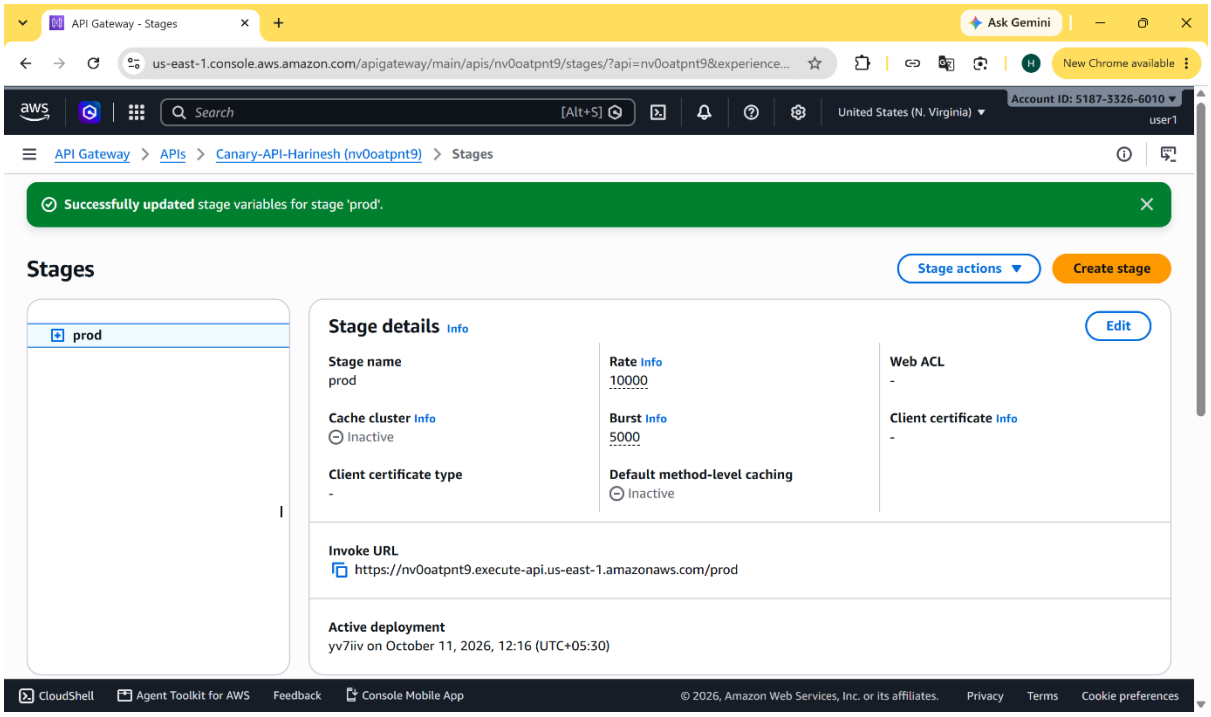


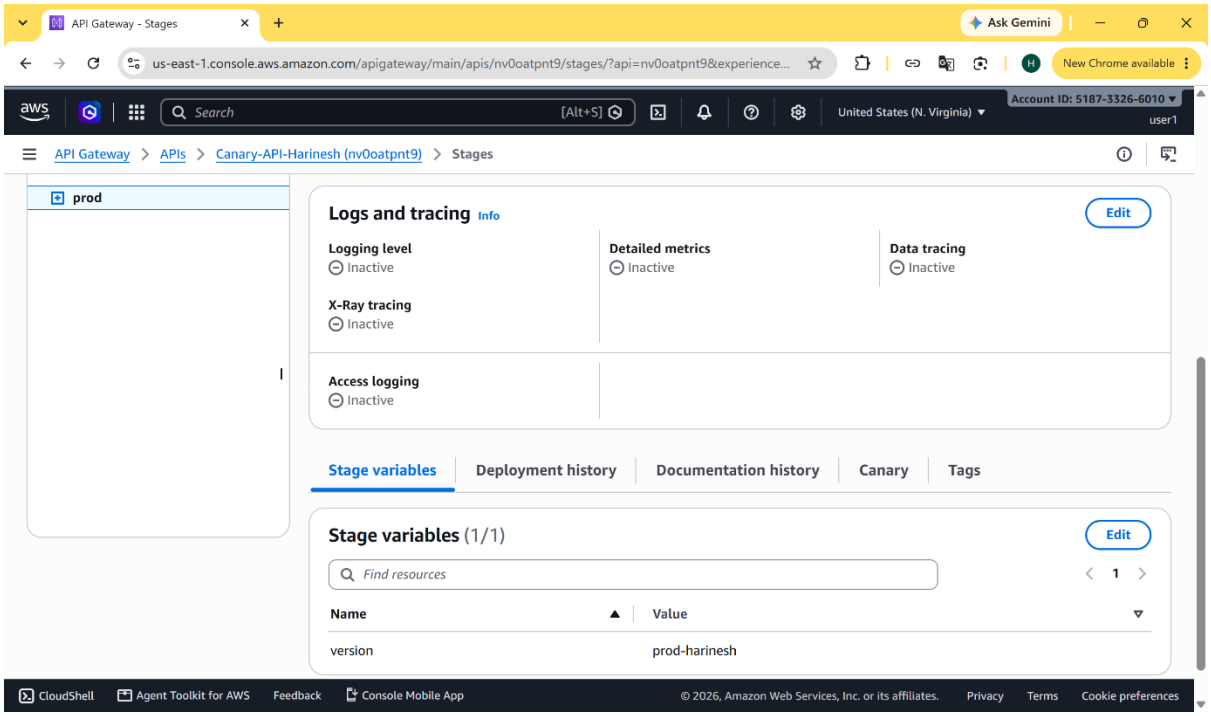



API Gateway - Create method
CloudShell | us-east-1
Ask Gemini
us-east-1.console.aws.amazon.com/cloudshell/home?region=us-east-1
New Chrome available
Search
[Alt+S]
United States (N. Virginia)
Account ID: 5187-3326-6010
user1
CloudShell
Actions
us-east-1
~ $ sudo su
bash-5.2# aws lambda add-permission --function-name "arn:aws:lambda:us-east-1:518733266010:function:canary-demo-harinesh:prod-harinesh" --source-arn "arn:aws:execute-api:us-east-1:518733266010:nv0oatpnt9/*/GET/getdata" --principal apigateway.amazonaws.com --statement-id b2b78c60-5ebd-4ae5-99b4-1284f4f28eef --action lambda:InvokeFunction
{
"Statement": "{\"Sid\":\"b2b78c60-5ebd-4ae5-99b4-1284f4f28eef\",\"Effect\":\"Allow\",\"Principal\":{\"Service\":\"apigateway.amazonaws.com\"},\"Action\":\"lambda:InvokeFunction\",\"Resource\":\"arn:aws:lambda:us-east-1:518733266010:function:canary-demo-harinesh:prod-harinesh\",\"Condition\":{\"ArnLike\":{\"AWS:SourceArn\":\"arn:aws:execute-api:us-east-1:518733266010:nv0oatpnt9/*/GET/getdata\"}}}"
}
bash-5.2#
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API Gateway - Resources
Ask Gemini
us-east-1.console.aws.amazon.com/apigateway/main/apis/nv0oatpnt9/resources/?api=nv0oatpnt9&ex...
New Chrome available
Search
[Alt+S]
United States (N. Virginia)
Account ID: 5187-3326-6010
user1
API Gateway > APIs > Resources - Canary-API-Harinesh (nv0oatpnt9)
Successfully created method 'GET' in 'getdata'. Redeploy your API for the update to take effect.
Resources
API actions
Deploy API
Create resource
/
/getdata
GET
Resource details
Update documentation
Enable CORS
Path
/
Resource ID
q444cy4xbd
Methods (0)
Delete
Create method
Method type
Integration type
Authorization
API key
No methods
No methods defined.
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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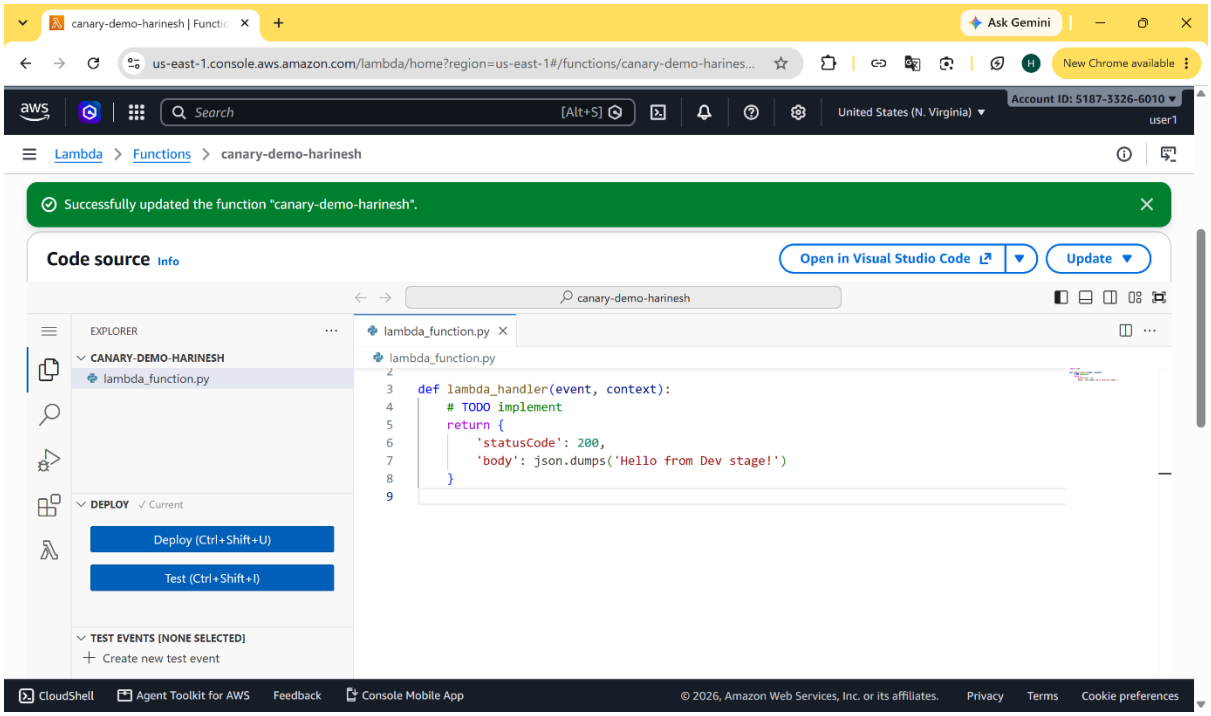

API Gateway - Stages
Ask Gemini
us-east-1.console.aws.amazon.com/apigateway/main/apis/nv0oatpnt9/stages?api=nv0oatpnt9&experience=...
New Chrome available
Search
[Alt+S]
United States (N. Virginia)
Account ID: 5187-3326-6010
user1
API Gateway > APIs > Canary-API-Harinesh (nv0oatpnt9) > Stages
Successfully created deployment for Canary-API-Harinesh. This deployment is active for prod.
Stages
Stage actions
Create stage
prod
Stage details Info
Edit
Stage name
prod
Rate Info
10000
Web ACL
-
Cache cluster Info
Inactive
Burst Info
5000
Client certificate Info
-
Client certificate type
-
Default method-level caching
Inactive
Invoke URL
https://nv0oatpnt9.execute-api.us-east-1.amazonaws.com/prod
Active deployment
yv7iiv on October 11, 2026, 12:16 (UTC+05:30)
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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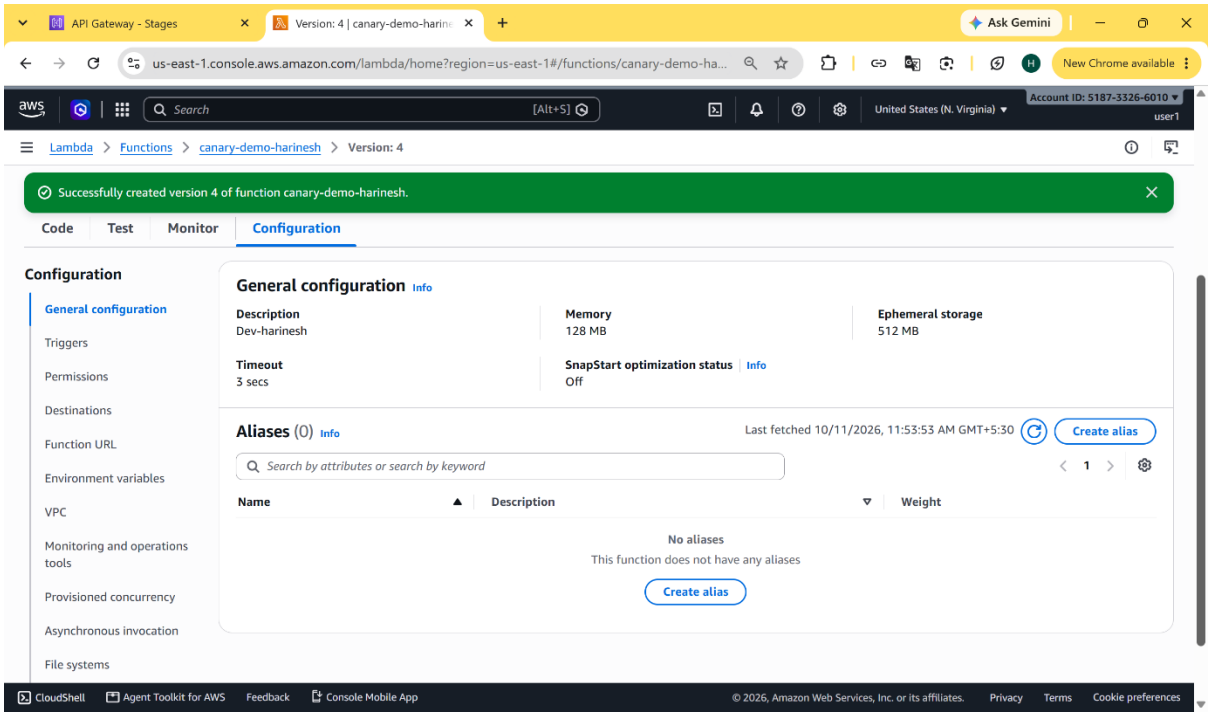

API Gateway - Stages
Ask Gemini
us-east-1.console.aws.amazon.com/apigateway/main/apis/nv0oatpnt9/stages/?api=nv0oatpnt9&experience...
New Chrome available
Search
[Alt+S]
United States (N. Virginia)
Account ID: 5187-3326-6010
user1
API Gateway > APIs > Canary-API-Harinesh (nv0oatpnt9) > Stages
Successfully updated stage variables for stage 'prod'.
Stages
Stage actions
Create stage
prod
Stage details Info
Edit
Stage name
prod
Rate Info
10000
Web ACL
-
Cache cluster Info
Inactive
Burst Info
5000
Client certificate Info
-
Client certificate type
-
Default method-level caching
Inactive
Invoke URL
https://nv0oatpnt9.execute-api.us-east-1.amazonaws.com/prod
Active deployment
yv7iiv on October 11, 2026, 12:16 (UTC+05:30)
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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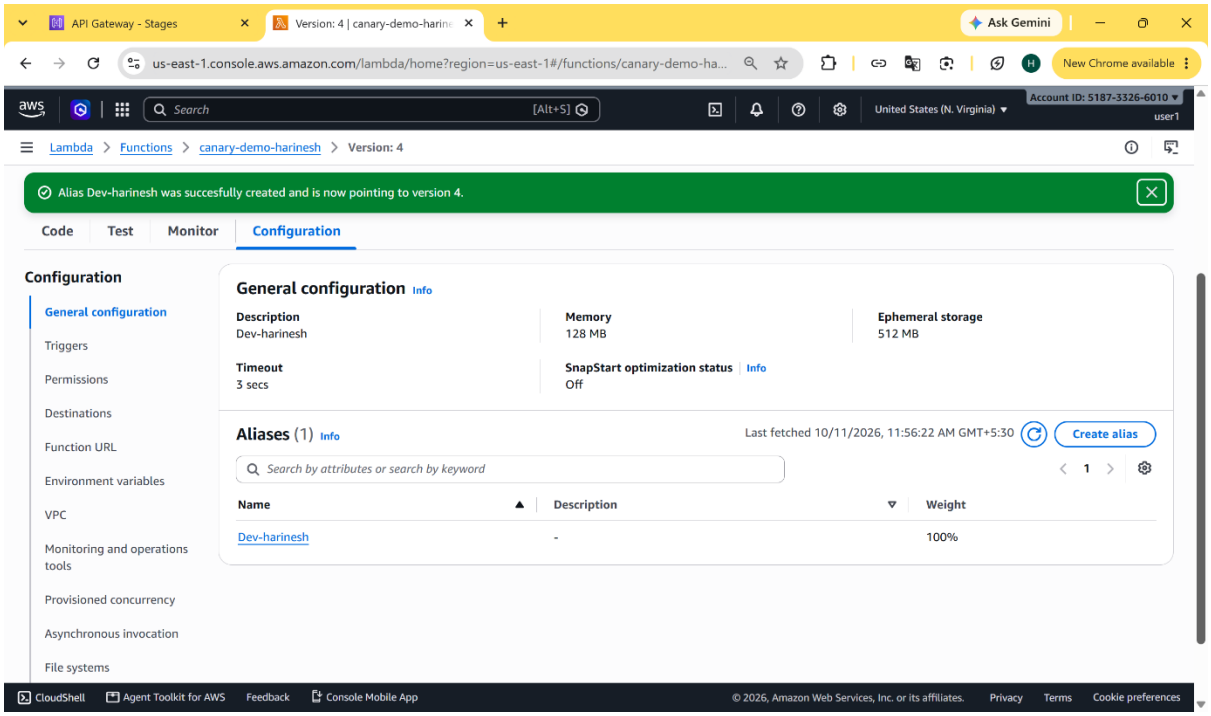


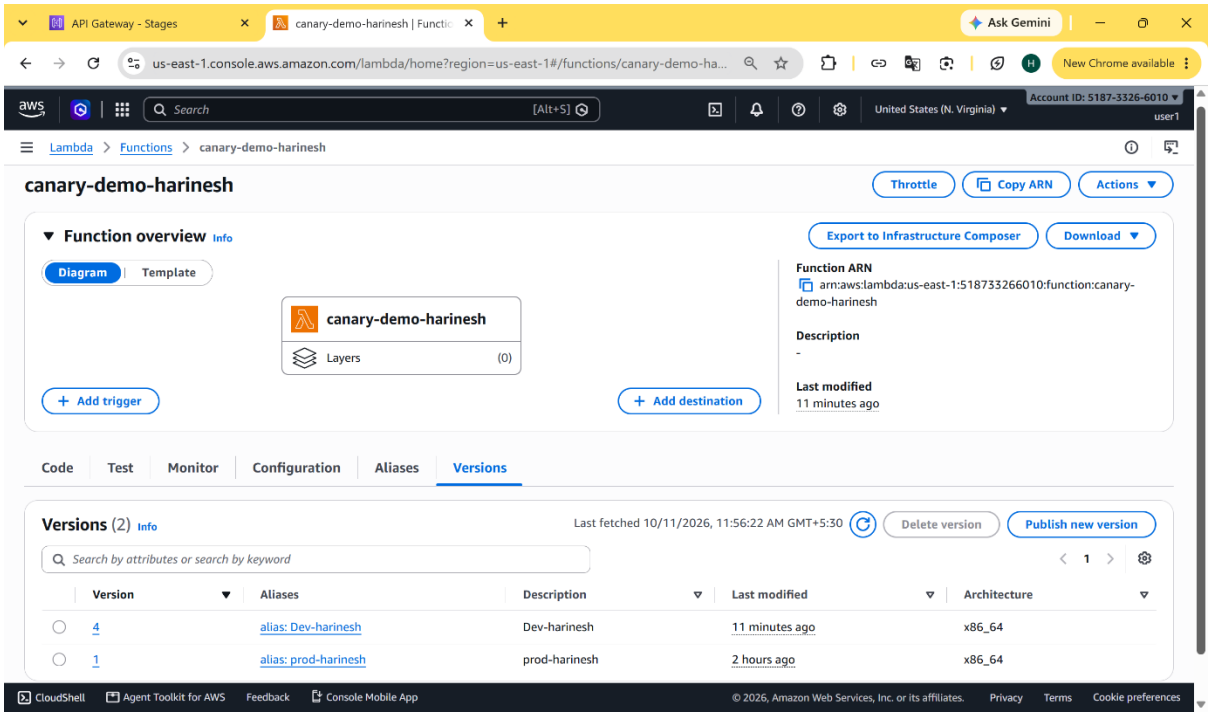


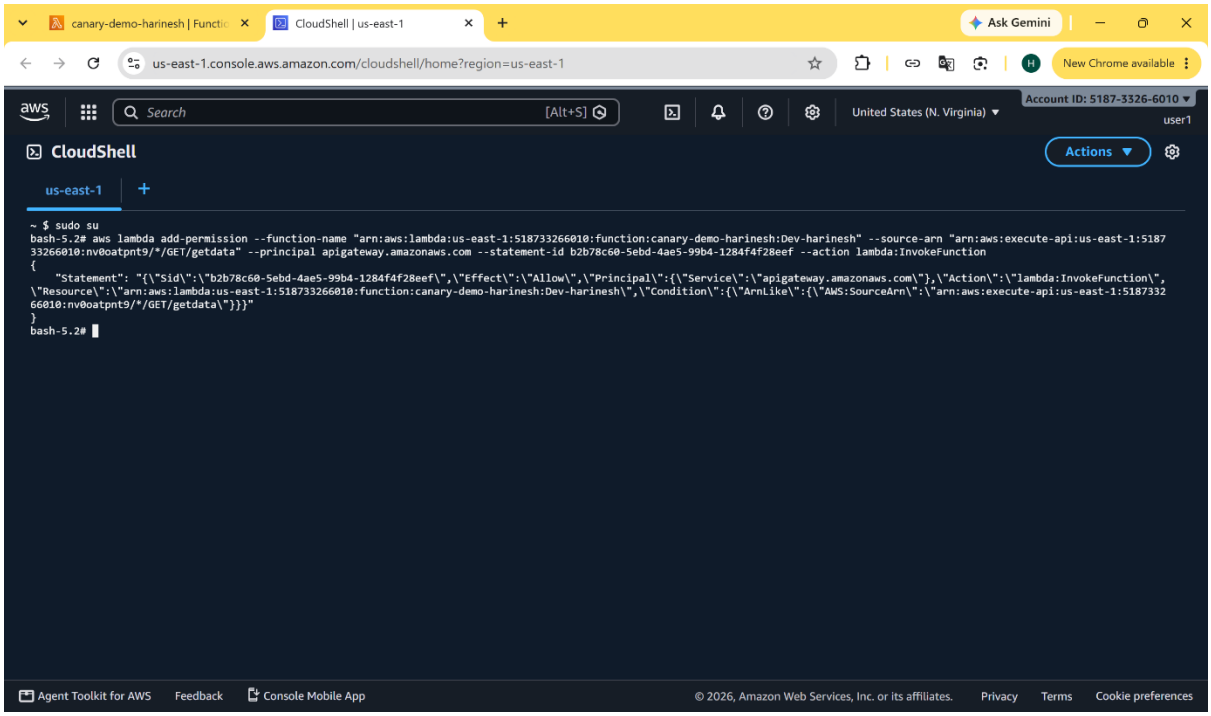
{"statusCode": 200, "body": "\"Hello from prod stage!\""}

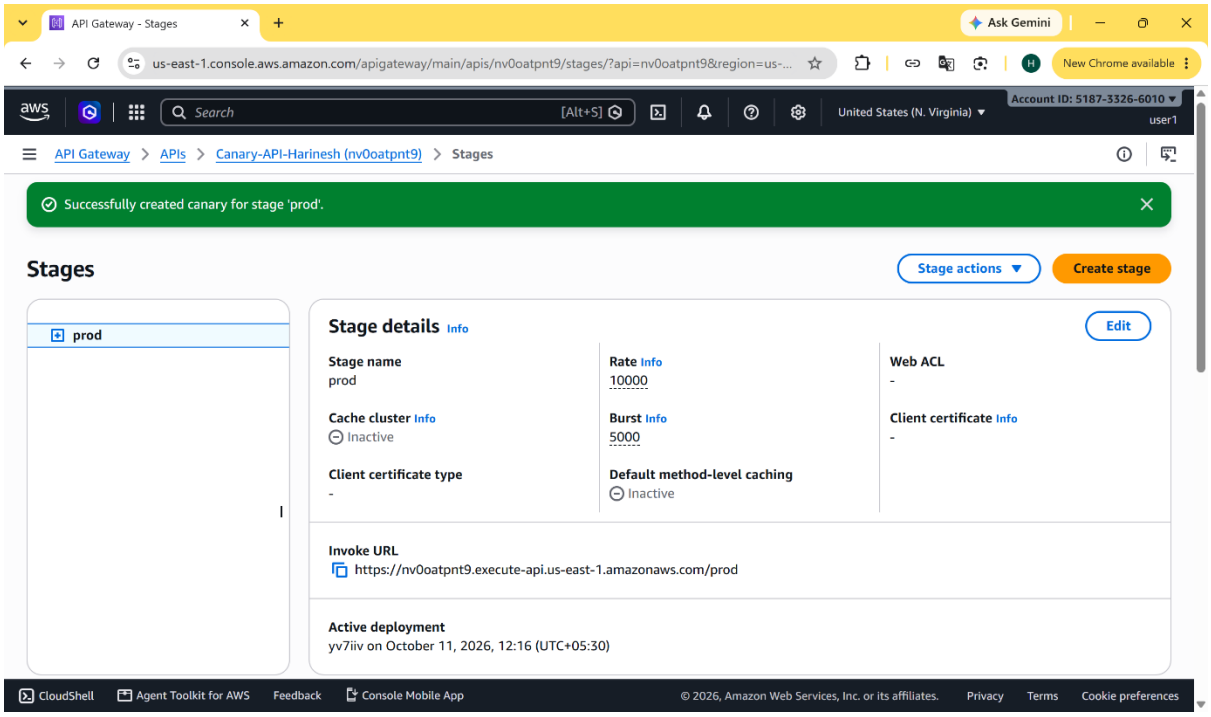

canary-demo-harinesh | Functic
Ask Gemini
us-east-1.console.aws.amazon.com/lambda/home?region=us-east-1#/functions/canary-demo-harines...
New Chrome available
Search
[Alt+S]
United States (N. Virginia)
Account ID: 5187-3326-6010
user1
Lambda > Functions > canary-demo-harinesh
Successfully updated the function "canary-demo-harinesh".
Code source Info
Open in Visual Studio Code
Update
canary-demo-harinesh
EXPLORER
CANARY-DEMO-HARINESH
lambda_function.py
lambda_function.py
def lambda_handler(event, context):
TODO implement
return {
'statusCode': 200,
'body': json.dumps('Hello from Dev stage!')
}
DEPLOY ✓ Current
Deploy (Ctrl+Shift+U)
Test (Ctrl+Shift+I)
TEST EVENTS [NONE SELECTED]
Create new test event
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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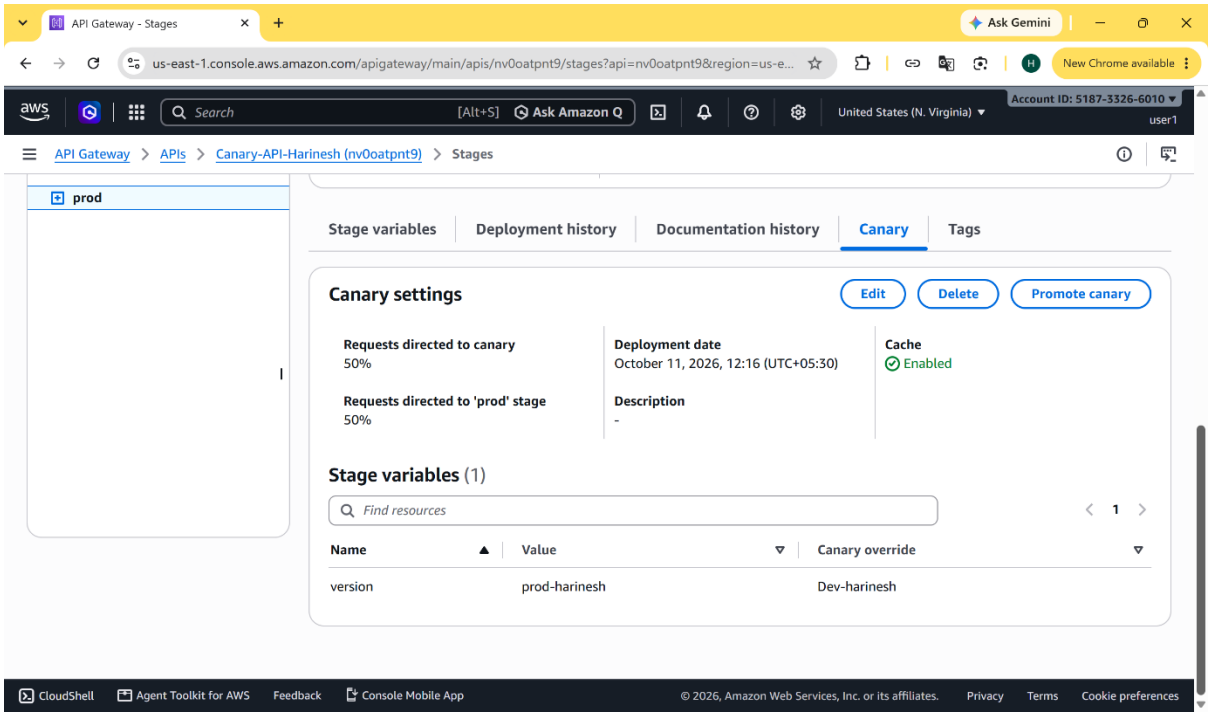

API Gateway - Stages
Version: 4 | canary-demo-harine
Ask Gemini
us-east-1.console.aws.amazon.com/lambda/home?region=us-east-1#/functions/canary-demo-ha...
New Chrome available
Search
[Alt+S]
United States (N. Virginia)
Account ID: 5187-3326-6010
user1
Lambda > Functions > canary-demo-harinesh > Version: 4
Successfully created version 4 of function canary-demo-harinesh.
Code
Test
Monitor
Configuration
Configuration
General configuration
Triggers
Permissions
Destinations
Function URL
Environment variables
VPC
Monitoring and operations tools
Provisioned concurrency
Asynchronous invocation
File systems
General configuration Info
Description
Dev-harinesh
Memory
128 MB
Ephemeral storage
512 MB
Timeout
3 secs
SnapStart optimization status Info
Off
Aliases (0) Info
Last fetched 10/11/2026, 11:53:53 AM GMT+5:30
Create alias
Search by attributes or search by keyword
Name
Description
Weight
No aliases
This function does not have any aliases
Create alias
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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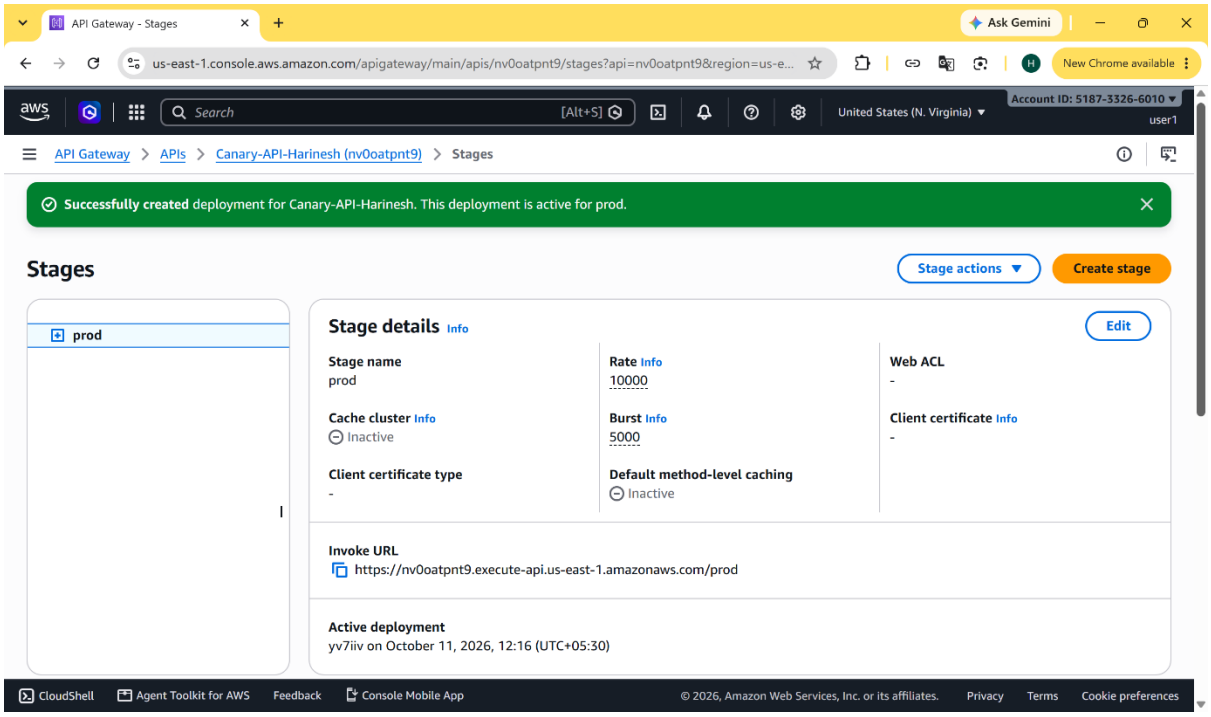

API Gateway - Stages
Version: 4 | canary-demo-harin
Ask Gemini
us-east-1.console.aws.amazon.com/lambda/home?region=us-east-1#/functions/canary-demo-ha...
New Chrome available
Search
[Alt+S]
United States (N. Virginia)
Account ID: 5187-3326-6010
user1
Lambda > Functions > canary-demo-harinesh > Version: 4
Alias Dev-harinesh was succesfully created and is now pointing to version 4.
Code
Test
Monitor
Configuration
Configuration
General configuration
Triggers
Permissions
Destinations
Function URL
Environment variables
VPC
Monitoring and operations tools
Provisioned concurrency
Asynchronous invocation
File systems
General configuration Info
Description
Dev-harinesh
Memory
128 MB
Ephemeral storage
512 MB
Timeout
3 secs
SnapStart optimization status Info
Off
Aliases (1) Info
Last fetched 10/11/2026, 11:56:22 AM GMT+5:30
Create alias
Search by attributes or search by keyword
Name
Description
Weight
Dev-harinesh
-
100%
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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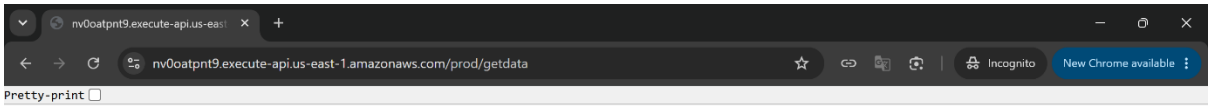

API Gateway - Stages
canary-demo-harinesh | Functic
Ask Gemini
us-east-1.console.aws.amazon.com/lambda/home?region=us-east-1#/functions/canary-demo-ha...
New Chrome available
Search
[Alt+S]
United States (N. Virginia)
Account ID: 5187-3326-6010
user1
Lambda > Functions > canary-demo-harinesh
canary-demo-harinesh
Throttle
Copy ARN
Actions
Function overview Info
Export to Infrastructure Composer
Download
Diagram
Template
canary-demo-harinesh
Layers
(0)
Add trigger
Add destination
Function ARN
arn:aws:lambda:us-east-1:518733266010:function:canary-demo-harinesh
Description
-
Last modified
11 minutes ago
Code
Test
Monitor
Configuration
Aliases
Versions
Versions (2) Info
Last fetched 10/11/2026, 11:56:22 AM GMT+5:30
Delete version
Publish new version
Search by attributes or search by keyword
Version
Aliases
Description
Last modified
Architecture
4
alias: Dev-harinesh
Dev-harinesh
11 minutes ago
x86_64
1
alias: prod-harinesh
prod-harinesh
2 hours ago
x86_64
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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canary-demo-harinesh | Functic
CloudShell | us-east-1
Ask Gemini
us-east-1.console.aws.amazon.com/cloudshell/home?region=us-east-1
New Chrome available
Search
[Alt+S]
United States (N. Virginia)
Account ID: 5187-3326-6010
user1
CloudShell
Actions
us-east-1
~ $ sudo su
bash-5.2# aws lambda add-permission --function-name "arn:aws:lambda:us-east-1:518733266010:function:canary-demo-harinesh:Dev-harinesh" --source-arn "arn:aws:execute-api:us-east-1:518733266010:nv0oatpnt9/*/GET/getdata" --principal apigateway.amazonaws.com --statement-id b2b78c60-5ebd-4ae5-99b4-1284f4f28eef --action lambda:InvokeFunction
{
"Statement": "{\"Sid\":\"b2b78c60-5ebd-4ae5-99b4-1284f4f28eef\",\"Effect\":\"Allow\",\"Principal\":{\"Service\":\"apigateway.amazonaws.com\"},\"Action\":\"lambda:InvokeFunction\",\"Resource\":\"arn:aws:lambda:us-east-1:518733266010:function:canary-demo-harinesh:Dev-harinesh\",\"Condition\":{\"ArnLike\":{\"AWS:SourceArn\":\"arn:aws:execute-api:us-east-1:518733266010:nv0oatpnt9/*/GET/getdata\"}}}"
}
bash-5.2#
Agent Toolkit for AWS
Feedback
Console Mobile App
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API Gateway - Stages
Ask Gemini
us-east-1.console.aws.amazon.com/apigateway/main/apis/nv0oatpnt9/stages/?api=nv0oatpnt9&region=us-...
New Chrome available
Search
[Alt+S]
United States (N. Virginia)
Account ID: 5187-3326-6010
user1
API Gateway > APIs > Canary-API-Harinesh (nv0oatpnt9) > Stages
Successfully created canary for stage 'prod'.
Stages
Stage actions
Create stage
prod
Stage details Info
Edit
Stage name
prod
Rate Info
10000
Web ACL
-
Cache cluster Info
Inactive
Burst Info
5000
Client certificate Info
-
Client certificate type
-
Default method-level caching
Inactive
Invoke URL
https://nv0oatpnt9.execute-api.us-east-1.amazonaws.com/prod
Active deployment
yv7iiv on October 11, 2026, 12:16 (UTC+05:30)
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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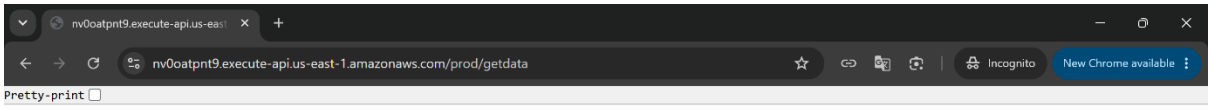

API Gateway - Stages
Ask Gemini
us-east-1.console.aws.amazon.com/apigateway/main/apis/nv0oatpnt9/stages?api=nv0oatpnt9&region=us-e...
New Chrome available
Search
[Alt+S]
Ask Amazon Q
United States (N. Virginia)
Account ID: 5187-3326-6010
user1
API Gateway > APIs > Canary-API-Harinesh (nv0oatpnt9) > Stages
prod
Stage variables
Deployment history
Documentation history
Canary
Tags
Canary settings
Edit
Delete
Promote canary
Requests directed to canary
50%
Requests directed to 'prod' stage
50%
Deployment date
October 11, 2026, 12:16 (UTC+05:30)
Description
-
Cache
Enabled
Stage variables (1)
Find resources
1
Name
Value
Canary override
version
prod-harinesh
Dev-harinesh
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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API Gateway - Stages
Ask Gemini
us-east-1.console.aws.amazon.com/apigateway/main/apis/nv0oatpnt9/stages?api=nv0oatpnt9&region=us-e...
New Chrome available
Search
[Alt+S]
United States (N. Virginia)
Account ID: 5187-3326-6010
user1
API Gateway > APIs > Canary-API-Harinesh (nv0oatpnt9) > Stages
Successfully created deployment for Canary-API-Harinesh. This deployment is active for prod.
Stages
Stage actions
Create stage
prod
Stage details Info
Edit
Stage name
prod
Rate Info
10000
Web ACL
-
Cache cluster Info
Inactive
Burst Info
5000
Client certificate Info
-
Client certificate type
-
Default method-level caching
Inactive
Invoke URL
https://nv0oatpnt9.execute-api.us-east-1.amazonaws.com/prod
Active deployment
yv7iiv on October 11, 2026, 12:16 (UTC+05:30)
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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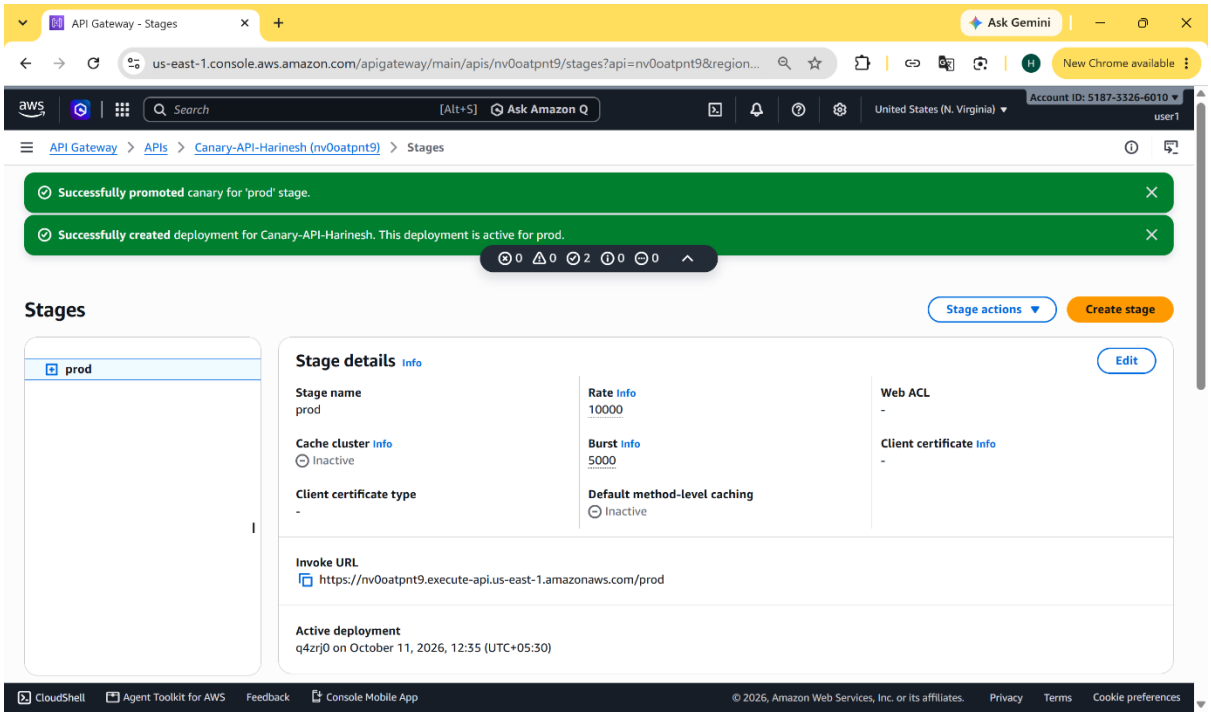
Pretty-print

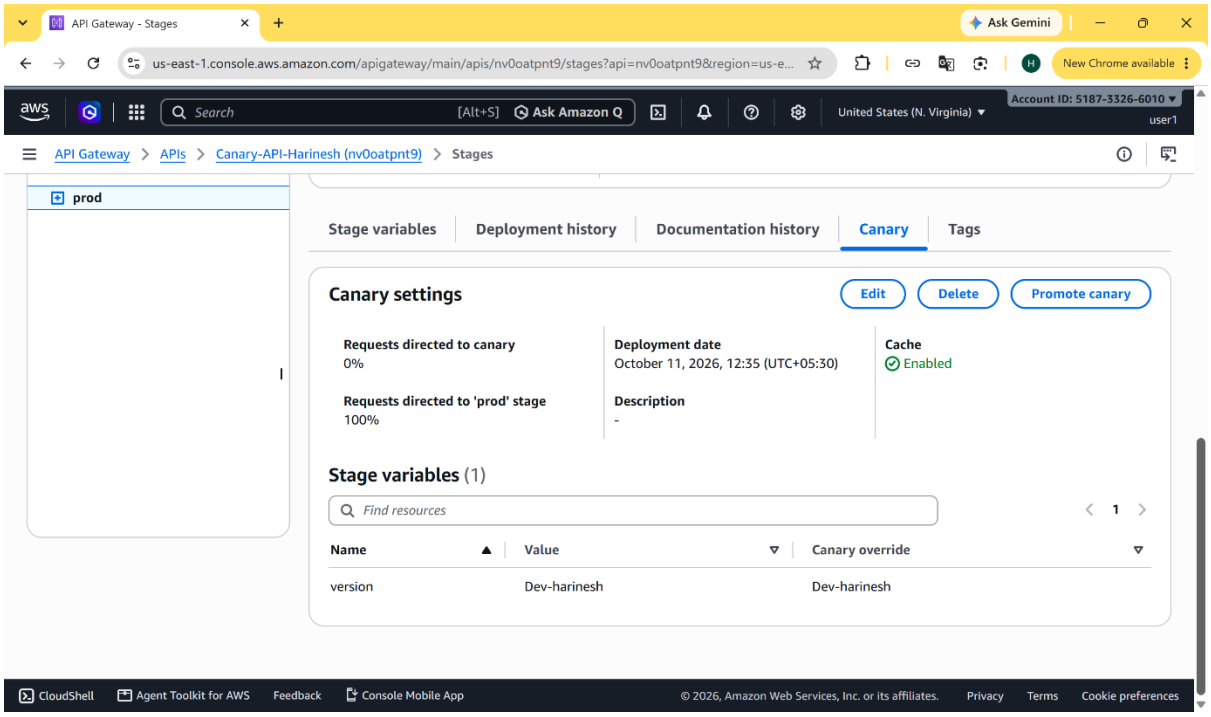
{"statusCode": 200, "body": "\"Hello from prod stage!\""}



Pretty-print

{"statusCode": 200, "body": "\"Hello from Dev stage!\""}


API Gateway - Stages
Ask Gemini
us-east-1.console.aws.amazon.com/apigateway/main/apis/nv0oatpnt9/stages?api=nv0oatpnt9&region...
New Chrome available
Search
[Alt+S]
Ask Amazon Q
United States (N. Virginia)
Account ID: 5187-3326-6010
user1
API Gateway > APIs > Canary-API-Harinesh (nv0oatpnt9) > Stages
Successfully promoted canary for 'prod' stage.
Successfully created deployment for Canary-API-Harinesh. This deployment is active for prod.
Stages
Stage actions
Create stage
prod
Stage details Info
Edit
Stage name
prod
Rate Info
10000
Web ACL
-
Cache cluster Info
Inactive
Burst Info
5000
Client certificate Info
-
Client certificate type
-
Default method-level caching
Inactive
Invoke URL
https://nv0oatpnt9.execute-api.us-east-1.amazonaws.com/prod
Active deployment
q4zrj0 on October 11, 2026, 12:35 (UTC+05:30)
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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API Gateway - Stages
Ask Gemini
us-east-1.console.aws.amazon.com/apigateway/main/apis/nv0oatpnt9/stages?api=nv0oatpnt9&region=us-e...
New Chrome available
Search
[Alt+S]
Ask Amazon Q
United States (N. Virginia)
Account ID: 5187-3326-6010
user1
API Gateway > APIs > Canary-API-Harinesh (nv0oatpnt9) > Stages
prod
Stage variables
Deployment history
Documentation history
Canary
Tags
Canary settings
Edit
Delete
Promote canary
Requests directed to canary
0%
Deployment date
October 11, 2026, 12:35 (UTC+05:30)
Cache
Enabled
Requests directed to 'prod' stage
100%
Description
-
Stage variables (1)
Find resources
1
Name
Value
Canary override
version
Dev-harinesh
Dev-harinesh
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
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Cookie preferences



Pretty-print

{"statusCode": 200, "body": "\"Hello from Dev stage!\""}